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Floods in India, Causes, Effects, Challenges & Measures - UPSC Notes

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Floods in India are a serious issue, with about 12% of the country's land being flood-prone, mostly in the Ganga-Brahmaputra regions. Floods are caused by heavy rain, dam failures, and poor urban planning. This leads to major impacts like the loss of lives and property (annual damage is around ₹1,800 crore). India manages this through flood forecasting (CWC), embankments, and a coordinated response from agencies like NDMA and IMD. Floods in places across the globe, like Texas, happen similarly due to excessive rainfall overwhelming river systems like the Guadalupe River.

This topic, Floods in India belongs to [General Studies Paper 3](#) (Disaster Management, Environment) for the UPSC IAS Examination.

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What Caused the July 2025 Flooding in Texas?

Global weather patterns often mirror each other. In July 2025, central Texas in the USA experienced a "1-in-1000-year" rain event. After a long period of drought, a **mesoscale convective vortex** (a rotating weather system) combined with tropical moisture to dump over 20 inches of rain in just a few hours. The Guadalupe River rose 32.5 feet in only 1.5 hours, leading to 137 deaths and over \$1.1 billion in damages (Source: NOAA). This event highlights how "weather whiplash" shifting from extreme dry to extreme wet is becoming a global challenge.



Floods in India

A flood is simply the overflow of water that covers land that is usually dry. In India, floods are a major and recurring problem affecting agriculture, infrastructure, and public health. Flooding can be minor or catastrophic, and it can happen quickly (flash flood) or build up slowly. Human changes to the environment, along with increased rainfall and extreme weather, often make floods worse and more frequent.



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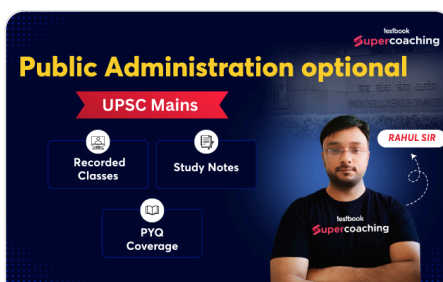


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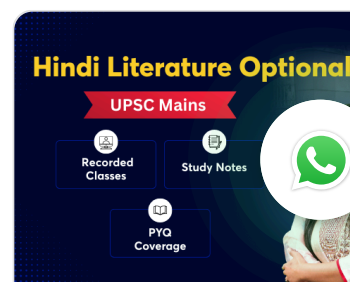


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What are the Flood Prone Areas in India?

India has a significant vulnerability to floods. Out of its total geographical area, about 40 million hectares are traditionally considered flood-prone, though recent estimates from the Ministry of Jal Shakti (2024) suggest this area has expanded to approximately 49.15 million hectares due to changing climate patterns.

The flood-prone plains are mainly distributed across four major regions:

- **Ganga River Region:** The lower and middle courses of the Ganga, including tributaries like the Kosi, Gandak, and Son, suffer from low gradients (flat plains) which slow drainage.
- **Brahmaputra River Region:** This basin, along with the Ganga and Meghna, accounts for about 60% of the country’s total river flow, making the region highly vulnerable, particularly Assam.
- **Central and Deccan Region:** The peninsular rivers such as Godavari and Krishna have shallow basins because of hard rock beds beneath them, so they get flooded easily during sudden heavy rainfalls.
- **North-West River Region:** Floods are common in the places along the Sutlej and Ravi rivers, especially in Punjab and Haryana, and the main cause is the drainage systems that are not sufficient to handle the situation.

State-wise studies state that maximum damage occurs in the Ganges-Brahmaputra-Meghna Basin. According to the Central Water Commission, Uttar Pradesh and Uttarakhand account for 33% of the flood damage, Bihar for 27%, and Punjab and Haryana for 15%.

Rivers that Generally Cause Floods in India

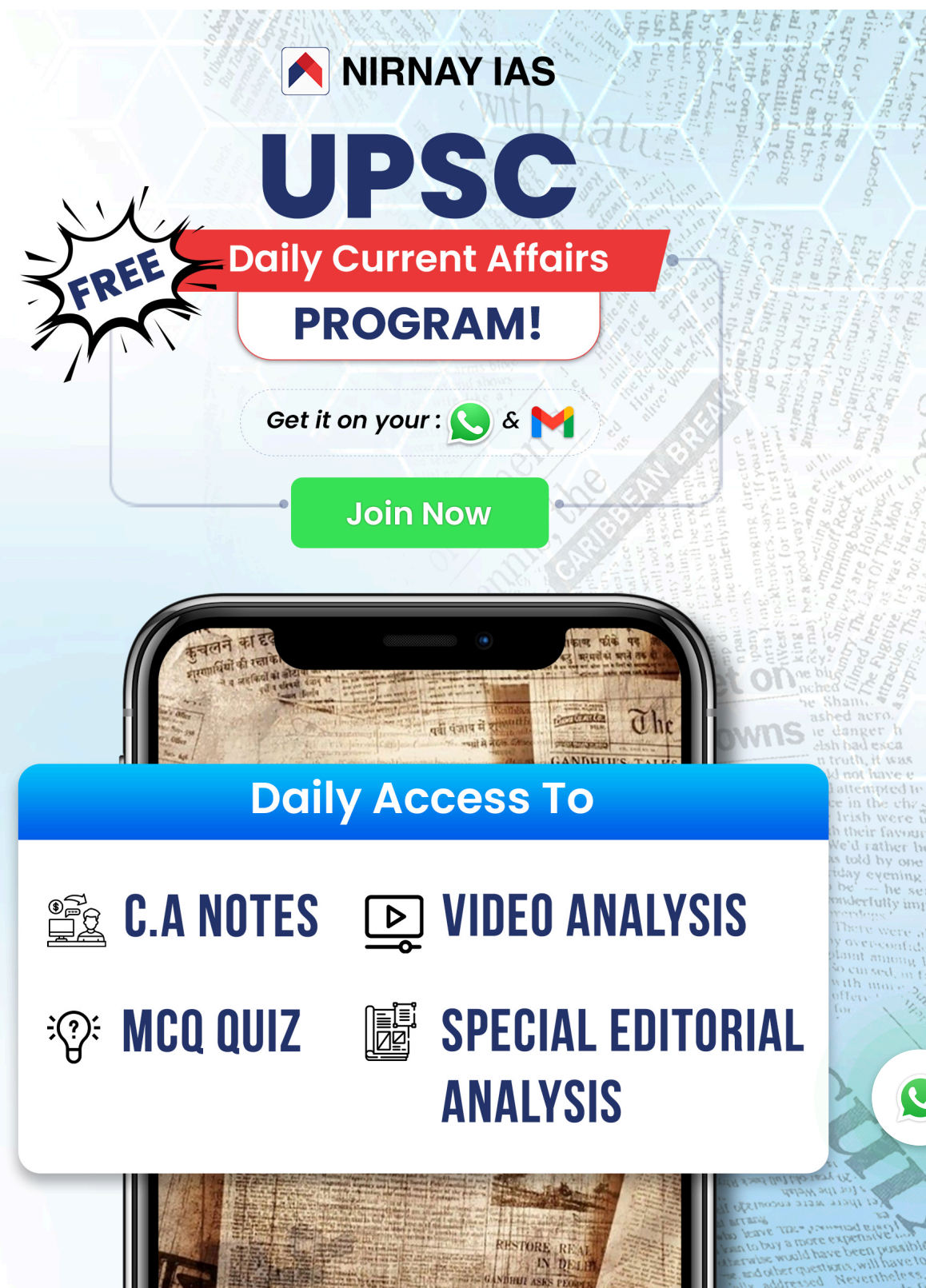
Different regions in India face threats from different river systems. The table below lists the primary rivers responsible for seasonal flooding:

River System	Major Rivers Involved	States Most Affected
Ganga Basin	Ganga, Kosi, Gandak, Ghaghra, Son	Bihar, Uttar Pradesh, West Bengal
Brahmaputra Basin	Brahmaputra, Barak, Lohit, Subansiri	Assam, Arunachal Pradesh



Peninsular Rivers	Godavari, Krishna, Mahanadi, Tapi	Odisha, Andhra Pradesh, Maharashtra
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Also read the article on [Urban Flooding in India](#) here!



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What are the Major Causes of Floods in India?

Floods result from a combination of natural weather events and human-made changes to the landscape. While nature sets the stage with seasonal rains, human actions often decide how much damage those rains will cause.

Natural Causes of Floods

Natural factors are largely driven by India's unique geography and the seasonal monsoon cycle.

Heavy Rain and Cloudbursts

This is the most frequent cause. During the monsoon, the ground often becomes fully soaked and cannot absorb any more water. This leads to high surface runoff, where water flows rapidly over the land into rivers or low-lying areas, often resulting in sudden "flash floods." Cloudbursts, which are extreme amounts of rain in a tiny window of time, are particularly common in the Himalayan foothills.

Overflowing of Rivers

When the volume of water exceeds what a river channel can hold, it spills over its banks. This is often made worse by **siltation**, where sand and soil settle at the bottom of the river, making it shallower and less able to carry large amounts of water. Rivers like the Kosi are notorious for changing their paths and overflowing during the peak monsoon.

Melting of Glaciers

In the Himalayan region, rapid melting of snow or ice can send a sudden wall of water downstream. **Glacial Lake Outburst Floods (GLOFs)** are a major concern in states like Uttarakhand and Sikkim, where natural dams holding glacial water can suddenly break without warning.

Cyclones and Storm Surges

Coastal states like Odisha, Andhra Pradesh, and West Bengal face floods from the sea. During a cyclone, strong winds push seawater onto the land. This "storm surge" can flood vast areas of salty water, destroying crops and homes. These events are most common during the post-monsoon months of October and November.

Anthropogenic (Man-Made) Causes of Floods

Human activities have altered the natural drainage of the land, making many regions more vulnerable to water accumulation.

Deforestation and Soil Erosion

Trees and plants act as natural sponges. When forests are cleared for timber or farming, there are no roots to hold the soil or slow down the rainwater. This leads to faster runoff and carries more silt in rivers, further increasing the risk of overflow and making flood events more intense.

Unplanned Urbanization

Cities are often built with concrete and asphalt, which are non-porous surfaces. Because the rain cannot seep into the ground, it stays on the surface. If the city's drainage pipes are old, blocked, or poorly designed, even a normal rainstorm can turn streets into rivers. This is why cities like Bengaluru and Mumbai face severe "urban flooding" almost every year.



water is released too suddenly during an emergency. Similarly, when an embankment (a protective wall along a river) breaks, it can lead to catastrophic flooding in the villages that thought they were protected.

Encroachment of Floodplains

People often build homes and factories right next to rivers in areas known as floodplains. When the river naturally expands during the rainy season, these structures block the water's path. This not only puts the people living there in danger but also forces the water into other areas that might not have flooded otherwise.

Check this article on the [West flowing rivers in India](#).

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What are the Major Impacts of Floods in India?

Floods are highly destructive events that cause extensive physical, economic, and psychological damage.

- **Loss of Lives:** This is the most serious effect. Fast-moving water can carry people away, and floodwaters can destroy buildings, creating unsafe environments. Floods are also responsible for the spread of waterborne diseases, which have deadly effects.
- **Damage to Property:** Floods cause enormous loss of belongings and property. Even two feet of water can wash a car away, and floodwaters destroy doors, windows, foundations, and corrode structural walls of buildings.
- **Economic Impacts:** Floods disrupt power plants, communication towers, roads, bridges, and water supply lines, requiring costly and time-consuming repairs. Business activities stop completely,



- **Psychosocial Effects:** Floods often result in lasting trauma. Losing loved ones, homes, or one's livelihood creates deep emotional stress and lasting psychological impacts, especially on children and displaced victims.

Read the article on [Disaster Management in India!](#)

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India's Preparedness for Floods

India manages flood risks through a system involving forecasting, construction, and disaster response agencies.

- **Flood Forecasting:** The Central Water Commission (CWC) runs a network of 175 flood forecasting stations across flood-prone states and UTs. This system, which started in 1985, issues daily flood warnings during the flood season (May to October) to help save lives and movable belongings.
- **Dam Construction:** Many large multi-purpose projects and dams were built during the Five-Year Plans to store surplus water. These structures help reduce runoff and release water under controlled conditions.
- **Embankment Construction:** Both Central and State governments have built numerous embankments along major rivers like the Brahmaputra, Kosi, and Godavari to prevent river water from overflowing onto agricultural and residential lands.
- **Run-Off Reduction:** This measure focuses on increasing the ground's ability to absorb surface water. This is achieved through large-scale afforestation (planting trees), particularly in the upper catchment areas, which slows down runoff.

Read the article on [Disaster Risk Reduction \(DRR\)!](#)

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What are the Challenges in Flood Management in India?

Effective flood control in India faces several major obstacles:

- **Trans-Boundary Rivers:** Many major rivers like the Ganga and Brahmaputra originate outside India, making it difficult to control the release of water or coordinate flood control measures with neighboring countries.
- **Lack of Drainage Maintenance:** River channels often have a low gradient and suffer from excessive siltation (accumulation of silt), which reduces their water-carrying capacity and causes them to overflow easily.
- **Encroachment:** Increasing human cities and building activities on flood plains (the areas near river) impede the free passage of water and amplify the damage during the flood.
- **Urban Drainage Failure:** The cities' old and badly taken care of storm water drainage systems are unable to manage the heavy and sudden rains, resulting in serious urban flooding.
- **Lack of Integrated Policy:** There is very often a lack of long-term, coordinated policy planning among different states and various agencies, which leads to incomplete or even contradictory flood control measures.



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What are the Measures to Control the Risk of Floods in India?

India manages flood risks through a mix of high-tech forecasting and physical construction.

Flood Forecasting and Warning

The **Central Water Commission (CWC)** operates a massive network of over 175 forecasting stations. They use satellite data and river sensors to predict when water levels will cross the "Danger Mark." This gives local authorities time to evacuate people. In 2024, the "FloodWatch India" app was upgraded to provide real-time alerts to citizens' mobile phones.

Structural Measures: Dams and Embankments

Dams are used to store excess water during heavy rain and release it slowly when the river can handle it. Embankments (raised walls along the riverbanks) are built to keep the water within the channel. However, these require constant maintenance to ensure they do not breach or break under pressure.

Nature-Based Solutions: Afforestation

Planting trees in the upper reaches of rivers (the catchment areas) is one of the best long-term solutions. Forests slow down the flow of rainwater and prevent soil from washing into the rivers, keeping the riverbeds deep and clear.

Urban Flood Management

The government has initiated the **Urban Flood Risk Management Programme**. In 2025, among the other major cities, Bhopal, Patna, and Guwahati received the approval of Phase 2 with a total of more than ₹2,444 crore budget (Source: PIB, 2025). The main point of the program is the drainage cleaning, sponge city infrastructure development, and the preservation of local lakes that function as natural reservoirs for stormwater.

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